

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 2.1: Similarity and Enlargement — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Driving Question	DRIVING QUESTION: Why does resizing a passport photo change its size but never its shape — and what is the mathematics that makes this possible?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Passport Photo Puzzle: Launching the Phenomenon	Two passport photos of the same image: one measuring 35 mm x 45 mm and one measuring 105 mm x 135 mm. Both show the same face with the same proportions. A third distorted photo (width tripled, height unchanged) in which the face looks squashed and unnatural.	When we divided the large photo measurements by the small photo measurements, we got the same number (3) for both width and height. The shape of the face is preserved only when both dimensions are scaled by the same number. The number that connects the two sizes is called the Linear Scale Factor.	We cannot yet fully explain why multiplying by the same number preserves shape, or what happens to the area or volume when we scale. These questions are recorded on the Driving Question Board and will be answered in the lessons that follow.
Lesson 2 The Passport Photo Puzzle — Discovering the Linear Scale Factor	Students measured five pairs of corresponding lengths on two passport photo prints (35 mm x 45 mm and 105 mm x 135 mm) and computed the ratio of each pair.	Every pair of corresponding lengths shares the same ratio — the Linear Scale Factor ($LSF = 3$) — meaning the photographer's software multiplied every length by exactly 3, which is why the shape is perfectly preserved even though the size tripled.	The LSF is calculated as $LSF = (\text{length in enlarged figure}) \div (\text{corresponding length in original figure})$. When the same LSF applies to ALL corresponding lengths, the shape is preserved and the two figures are mathematically similar.
Lesson 3 Identifying Properties of Similar Figures: Angles, Ratios, and the Rule Behind the Resize	Learners measured corresponding angles (using protractors) and corresponding side lengths (using rulers) in three pairs of rectangles based on Nairobi passport photo dimensions and two pairs of triangles. They computed ratios of corresponding sides as decimals to 2 d.p. and recorded all findings in a structured Evidence Table. They conducted a Gallery Walk to compare group data using a dot-voting system.	Two figures are similar if and only if: (1) all corresponding angles are equal, AND (2) all ratios of corresponding sides are equal (this constant ratio is the Linear Scale Factor). Changing only one dimension — as in the distorted Pair C rectangle — destroys similarity because the side ratios become unequal, even though the angles may remain equal in a rectangle. Similarity is a multiplicative relationship, not an additive	Learners explained their Evidence Table findings to peers during the Gallery Walk and to the whole class during the Explain phase, justifying 'similar' or 'not similar' verdicts using specific ratio values. They applied the Similarity Checklist to three new Kenya-relevant contexts (road-sign triangles, a cooking pot, and a WhatsApp profile photo crop), wrote 2-sentence justifications using ratio language, and

		one.	revised their cumulative similarity model diagrams to show both conditions with annotated measurements.
Lesson 4 Discovering Similar Figures: Equal Angles and Constant Ratios	Learners measured corresponding sides and angles of two pairs of polygon cut-outs (a rectangle pair from a Kenyan newspaper advertisement and a triangle pair from a Kenyan road-map extract). They recorded all measurements in a structured table and computed the ratio of each pair of corresponding sides.	Learners discovered that in both figure pairs: (1) all corresponding angles were equal, and (2) all ratios of corresponding sides were equal (constant), equalling the linear scale factor. They formalised these two findings as the definition of similar figures and applied the definition to classify new figure pairs as similar or non-similar.	The two defining properties of similarity — equal corresponding angles and constant ratios of corresponding sides — explain why the passport photo looks identical at two different sizes: the photographer's software applied one linear scale factor (3) to every length, preserving all angles and keeping all side ratios equal. When only width is scaled (distorted photo), the ratio condition is violated and the image loses its shape.
Lesson 5 Linear Scale Factor and Lengths — Explaining the Photographer's Rule	Learners observed that when corresponding lengths from multiple pairs of similar figures are divided, they consistently produce the same constant ratio — regardless of which pair of corresponding sides is chosen. This pattern held across rectangles (passport photos), triangles (rooftop trusses, pennants, road signs), and other polygons (tiles, kite frames, land plots).	Learners formalised this observed pattern as the Linear Scale Factor (LSF): the single constant ratio of any corresponding length in the image to the corresponding length in the original. They learned to calculate LSF, use it to find unknown lengths in similar figures, and verify similarity by checking consistency of LSF across multiple pairs of corresponding sides.	Learners explained — using Claim–Evidence–Reasoning — why the photographer's software must apply the SAME LSF to BOTH the width and height of the passport photo. When only one dimension is scaled (the photographer's mistake), the two 'ratios' differ, the figure is no longer similar to the original, and the face appears distorted. The $LSF = 3$ connecting the 35×45 mm print to the 105×135 mm print is the mathematical rule that preserves shape while changing size.
Lesson 6 Calculating and Applying the Linear Scale Factor (LSF) to Find Unknown Lengths	Students revisited their own Lesson 2 measurement data (passport photo: $35 \text{ mm} \times 45 \text{ mm}$ enlarged to $105 \text{ mm} \times 135 \text{ mm}$) and calculated width ratio, height ratio, and diagonal ratio, finding all three equal to 3. They observed a teacher think-aloud worked example of the Kimathi Street passport photo problem and a school badge problem from Nakuru, watching how a single pair of corresponding lengths unlocks all unknowns.	Students learned that the constant ratio of any image length to its corresponding object length is called the Linear Scale Factor (LSF), calculated as $LSF = \text{image length} \div \text{object length}$. They learned that $LSF > 1$ indicates enlargement, $LSF < 1$ indicates reduction, and $LSF = 1$ indicates congruence. For the passport photo, $LSF = 105 \div 35 = 3$, meaning every length is multiplied by 3. They also learned that applying different LSFs to different dimensions (e.g., width $\times 3$ but height $\times 2$) destroys similarity and distorts the image.	Students explained, through DQB sentence completion and cumulative model annotation, that the photographer's software preserves the face perfectly because it multiplies every length — width, height, and all internal distances — by the same LSF ($= 3$). Because every length scales by the same factor, all ratios between lengths remain unchanged, all angles remain unchanged, and the shape is perfectly preserved. This is the mathematical rule that makes resizing without distortion possible.
Lesson 7 Area Scale Factor — How Does Enlargement Change the Area of the Passport Photo?	Learners measured the dimensions of the standard passport photo ($35 \text{ mm} \times 45 \text{ mm}$) and the enlarged passport photo ($105 \text{ mm} \times 135 \text{ mm}$), calculated the area of each, and computed the ratio of areas. They repeated	Learners derived and confirmed that the Area Scale Factor (ASF) equals the square of the Linear Scale Factor (LSF): $ASF = LSF^2$. For the passport photos ($LSF = 3$), $ASF = 9$ — the enlarged photo uses 9 times	Learners applied $ASF = LSF^2$ to two real-world problems: a maize farm plot in Kericho ($LSF = 5$, area = $3,000 \text{ m}^2$) and a road sign near Nakuru ($LSF = 4$, area = $14,400 \text{ cm}^2$). They updated their cumulative

	this for a pair of similar Kenyan flag rectangles ($LSF = 2$). In both cases they recorded the ratio of areas in a structured ratio table.	more paper area, not 3 times. This relationship was proved algebraically: if sides scale by k , area scales by k^2 because $Area = (ka)(kb) = k^2(ab)$.	Similarity Models with area labels and the annotation $ASF = LSF^2$, and posted new DQB notes asking what happens to volume when a 3D object is enlarged — setting up the Volume Scale Factor investigation in Lesson 8.
Lesson 8 Area Scale Factor: Why Does the Larger Passport Photo Use Nine Times as Much Paper?	Learners drew similar rectangles (Kenyan farm plots) with $LSF = 2, 3$, and 4 on graph paper, counted grid squares to find areas, and recorded results in a class data table showing area ratios of $4, 9$, and 16 respectively.	For any two similar figures with linear scale factor k , the Area Scale Factor is k^2 ($ASF = LSF^2$). This means the enlarged passport photo ($LSF = 3$) has $3^2 = 9$ times the area of the standard print, which is why it uses significantly more paper and costs more to produce.	The squaring relationship arises because area is two-dimensional: when both length and width are each multiplied by k , the area is multiplied by $k \times k = k^2$. An algebraic proof for a general rectangle ($l \times w$ scaled to $kl \times kw$) confirms the pattern discovered from the graph-paper data.
Lesson 9 Locating the Centre of Enlargement: Where Was the Camera?	When we drew straight lines through each pair of corresponding vertices of the small passport photo and the large noticeboard print, all three lines met at exactly one point, which we called the centre of enlargement. The ratio of the distance from the centre to the image point divided by the distance from the centre to the object point equalled the linear scale factor (3 for the passport photo).	The centre of enlargement is the unique point through which all lines joining corresponding object and image points pass (collinearity property). Given any object and its enlarged or reduced image, we can locate the centre by construction (drawing lines through corresponding points and finding their intersection) or by algebra (finding the intersection of two such lines using coordinate geometry). This centre corresponds to the position of the camera lens or projector in real-world enlargement situations.	The passport photographer's software resized the image by applying a single linear scale factor ($LSF = 3$) from a fixed centre point — equivalent to the camera lens position. Every point in the original photo was moved along the straight line from the centre through that point, to a new position exactly 3 times as far from the centre. This is why the shape is perfectly preserved: all angles remain equal and all length ratios remain constant. By knowing the small print and the large print, we can reverse-engineer exactly where the camera was positioned, using the collinearity property of the centre of enlargement.
Lesson 10 MODEL BUILDING: Constructing a Scale Drawing of Our School Space	We observed that when we applied one linear scale factor to every dimension of our classroom, the resulting floor plan had exactly the same shape as the real room — all angles were preserved and all length ratios were constant — just as the passport photo studio software preserved the shape of the students face by scaling every pixel distance by the same factor of 3 .	We learned that a scale drawing is a real-life example of a similar figure: the linear scale factor (written as $1 : n$) controls every length uniformly, so any real space can be reduced to fit on paper without distortion, and the area of the drawing relates to the actual area by the square of the LSF .	We explained that the passport photo principle — one number scales everything equally — works at every physical scale, from a 35 mm photograph to a 12 -metre classroom wall. This is why architects, cartographers, and photographers all use the same mathematical idea: choose one LSF , apply it to every length, and the shape of the object is perfectly preserved in the new size.
Lesson 11 EXPLAIN & APPLY: Mixed Problems — LSF, ASF, and VSF	Learners observed three fully worked examples (Kimathi Street passport photo area, Kisumu githeri pot height ratio, Eldoret maize silo surface area) that modelled the complete $LSF \rightarrow ASF \rightarrow VSF$ chain, and completed four mixed Kenyan-context	Learners consolidated that LSF , $ASF (= LSF^2)$, and $VSF (= LSF^3)$ form a single unified system; that diagnosing which scale factor a problem provides — and which it demands — is the critical first step; and that the passport photographer's software	Learners explained in writing why the resized passport photo looks identical but occupies a different area: the LSF scales every length equally (preserving all proportions and angles), the area changes by LSF^2 , and if the photo were three-

	problems independently and in pairs.	preserves shape because every length is multiplied by the same LSF, keeping all angle ratios and length ratios constant.	dimensional the volume would change by LSF^3 — demonstrating that one number, the Linear Scale Factor, governs the entire transformation.
Lesson 12 Final Explanation — Weaving the Three Relationships: LSF, ASF, and VSF	Learners examined three Kenyan-context problems (Nairobi–Nakuru map, Mombasa architect's roof model, Kericho tea-farm water tanks) that each required a different scale-factor relationship — linear, area, or volume — and peer-marked each other's solutions using a three-colour protocol.	$ASF = LSF^2$ and $VSF = LSF^3$: a single Linear Scale Factor simultaneously determines how lengths, areas, and volumes change between similar figures. All corresponding angles remain equal and all corresponding lengths remain in the same ratio — these are the two conditions for similarity. If only one dimension is scaled (unequal LSF), angles change and the figure is no longer similar (distortion occurs).	The passport photo photographer's software applied one $LSF = 3$ equally to both width and height of the digital image. This preserved all angles (so the face looks identical) and all length ratios (so proportions are perfect), while making the area exactly $9\times$ larger ($ASF = 3^2 = 9$) and — if the image were a 3D object — the volume would be $27\times$ larger ($VSF = 3^3 = 27$). The mathematics that makes perfect resizing possible is the Linear Scale Factor applied uniformly to every dimension.

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